Daffodil International University
Department of Software Engineering
Faculty of Science & Information Technology
Final Examination, Summer 2025
Course Code: ENG 101; Course Title: English-1
Sections & Teachers: (A-N); MA, JC, MAR, SASI.

Time: 2:00 Hrs **Marks: 40**

Answer ALL Questions

[The figures in the right margin indicate the full marks and corresponding course outcomes. All portions of each question must be answered sequentially.]

Read the passage carefully and answer the following questions.

Accidental Scientists

A. A paradox lies close to the heart of scientific discovery. If you know just what you are looking for, finding it can hardly count as a discovery, since it was fully anticipated. But if, on the other hand, you have no notion of what you are looking for, you cannot know when you have found it, and discovery, as such, is out of the question. In the philosophy of science, these extremes map onto the purist forms of deductivism and inductivism: In the former, the outcome is supposed to be logically contained in the premises you start with; in the latter, you are recommended to start with no expectations whatsoever and see what turns up.

B. As in so many things, the ideal position is widely supposed to reside somewhere in between these two impossible-to-realise extremes. You want to have a good enough idea of what you are looking for to be surprised when you find something else of value, and you want to be ignorant enough of your end point that you can entertain alternative outcomes. Scientific discovery should, therefore, have an accidental aspect, but not too much of one. Serendipity is a word that expresses a position something like that. It's a fascinating word, and the late Robert King Merton—"the father of the sociology of science"—liked it well enough to compose its biography, assisted by the French cultural historian Elinor Barber.

C. The word did not appear in the published literature until the early 19th century and did not become well enough known to use without explanation until sometime in the first third of the 20th century. Serendipity means a "happy accident" or "pleasant surprise", specifically, the accident of finding something good or useful without looking for it. The first noted use of "serendipity" in the English language was by Horace Walpole. He explained that it came from the fairy tale, called *The Three Princes of Serendip* (the ancient name for Ceylon, or present day Sri Lanka), whose heroes "were always making discoveries, by accidents and sagacity, of things which they were not in quest of".

D. Antiquarians, following Walpole, found use for it, as they were always rummaging about for curiosities, and unexpected but pleasant surprises were not unknown to them. Some people just seemed to have a knack for that sort of thing, and serendipity was used to express that special capacity. The other community that came to dwell on serendipity to say something important about their practice was that of scientists, and here usages cut to the heart of the matter and were often vigorously contested. Many scientists, including the Harvard physiologist Walter Cannon and, later, the British immunologist Peter Medawar, liked to emphasise how much of scientific discovery was unplanned and even accidental. One of the examples is Hans Christian Ørsted's discovery of electromagnetism when he unintentionally brought a current-carrying wire parallel to a magnetic needle. The rhetoric about the sufficiency of the rational method was so hot. Indeed, as Medawar insisted, "There is no such thing as The Scientific Method," no way at all of systematising the process of discovery. Really important discoveries had a way of showing up when they had a mind to do so and not when you were looking for them. Maybe some scientists, like some book collectors, had a happy knack; maybe serendipity described the situation rather than a personal skill or capacity.

E. Some scientists using the word meant to stress those accidents belonging to the situation; some treated serendipity as a personal capacity; many others exploited the ambiguity of the notion. Yet what Cannon and Medawar took as a benign nose-thumbing at Dreams of Method, other scientists found incendiary. To say that science had a significant serendipitous aspect was taken by some as dangerous denigration. If scientific discovery were really accidental, then what was the special basis of expert authority? In this connection, the aphorism of choice came from no less an authority on scientific discovery than Louis Pasteur: "Chance favors the prepared mind." Accidents may happen, and things may turn up unplanned and unforeseen, as one is looking for something else, but the ability to notice such events, to see potential bearing and meaning, to exploit their occurrence and make constructive use of them—these are the results of systematic mental preparation. What seems like an accident is just another form of expertise. On closer inspection, it is insisted, the accident dissolves into sagacity.

F. For Merton himself—who one supposes must have been the senior author – serendipity represented the keystone in the arch of his social scientific work. In 1936, as a very young man, Merton wrote a seminal essay on "The Unanticipated Consequences of Purposive Social Action." It is, he argued, the nature of social action that what one intends is rarely what one gets: Intending to provide resources for buttressing Christian religion, the natural philosophers of the Scientific Revolution laid the groundwork for secularism; people wanting to be alone with nature in Yosemite Valley wind up crowding one another. We just don't know enough—and we can never know enough—to ensure that the past is an adequate guide to the future: Uncertainty about outcomes, even of our best-laid plans, is endemic. All social action, including that undertaken with the best evidence and formulated according to the most rational criteria, is uncertain in its consequences.

1.	***a.***	**Select** the most suitable headings for the given paragraphs from the list of ten headings below. Write the appropriate number **I-X** with paragraph list **i-v** in your script. There are more headings than the number of paragraphs, so you will not use them all. *For example: Paragraph G= IV* **List of headings:** i) Examples of some scientific discoveries ii) Varied details about what lies inside Horace Walpole's fairy tale iii) Resolving the contradiction iv) What is the Scientific Method v) The contradiction of views on scientific discovery vi) Some misunderstandings of serendipity vii) Opponents of authority viii) Reality doesn't always match expectation ix) How the word 'Serendipity" came into being x) Illustration of serendipity in the business sector. **1. Paragraph A** ________ **2. Paragraph C** ________ **3. Paragraph D** ________ **4. Paragraph E** ________ **5. Paragraph F** ________	[5x1=5]	**CLO-3** ***Level-C4***
	b)	**Identify** the correct answer to the following questions. **i.** In the passage, The word "inductivism" means- **A .** anticipate results in the beginning. **C.** accept chance discoveries **B.** work with prepared premises. **D.** look for what you want. **ii.** Medawar says "there is no such thing as The Scientific Method" because- **A.** discoveries are made by people with a determined mind. **B.** discoveries tend to happen unplanned. **C.** The process of discovery is unpleasant. **D.** Serendipity is not a skill. **iii.** Many scientists dislike the idea of serendipity because **A.** It is easily misunderstood and abused. **C.** it is too unpredictable. **B.** It is beyond their comprehension. **D.**it devalues their scientific expertise. **iv.** The writer mentions Irving Langmuir to illustrate **A.** planned science should be avoided. **B.** Industrial development needs uncertainty. **C.** people tend to misunderstand the relationship between cause and effect. **D.** Accepting uncertainty can help produce positive results. **V.** The example of Yosemite is to show- **A.** The conflict between reality and expectation. **B.** The importance of systematic planning.	[5x1=5]	

C. the intention of social action.
D. The power of anticipation.

Read the passage below and answer the following questions.

The Dead Sea Scrolls

In late 1946 or early 1947, three Bedouin teenagers were tending their goats and sheep near the ancient settlement of Qumran, located on the northwest shore of the Dead Sea in what is now known as the West Bank. One of these young shepherds tossed a rock into an opening on the side of a cliff and was surprised to hear a shattering sound. He and his companions later entered the cave and stumbled across a collection of large clay jars, seven of which contained scrolls with writing on them. The teenagers took the seven scrolls to a nearby town where they were sold for a small sum to a local antiquities dealer. Word of the find spread, and Bedouins and archaeologists eventually unearthed tens of thousands of additional scroll fragments from 10 nearby caves; together they make up between 800 and 900 manuscripts. It soon became clear that this was one of the greatest archaeological discoveries ever made.

The origin of the Dead Sea Scrolls, which were written around 2,000 years ago between 150 BCE and 70 CE, is still the subject of scholarly debate even today. According to the prevailing theory, they are the work of a population that inhabited the area until Roman troops destroyed the settlement around 70 CE. The area was known as Judea at that time, and the people are thought to have belonged to a group called the Essenes, a devout Jewish sect.

The majority of the texts on the Dead Sea Scrolls are in Hebrew, with some fragments written in an ancient version of its alphabet thought to have fallen out of use in the fifth century BCE. But there are other languages as well. Some scrolls are in Aramaic, the language spoken by many inhabitants of the region from the sixth century BCE to the siege of Jerusalem in 70 CE. In addition, several texts feature translations of the Hebrew Bible into Greek.

The Dead Sea Scrolls include fragments from every book of the Old Testament of the Bible except for the Book of Esther. The only entire book of the Hebrew Bible preserved among the manuscripts from Qumran is Isaiah; this copy, dated to the first century BCE, is considered the earliest biblical manuscript still in existence. Along with biblical texts, the scrolls include documents about sectarian regulations and religious writings that do not appear in the Old Testament.

The writing on the Dead Sea Scrolls is mostly in black or occasionally red ink, and the scrolls themselves are nearly all made of either parchment (animal skin) or an early form of paper called 'papyrus'. The only exception is the scroll numbered 3Q15, which was created out of a combination of copper and tin. Known as the Copper Scroll, this curious document features letters chiselled onto metal – perhaps, as some have theorized, to better withstand the passage of time. One of the most intriguing manuscripts from Qumran, this is a sort of ancient treasure map that lists dozens of gold and silver caches. Using an unconventional vocabulary and odd spelling, it describes 64 underground hiding places that supposedly contain riches buried for safekeeping. None of these hoards have been recovered, possibly because the Romans pillaged Judea during the first century CE. According to various hypotheses, the treasure belonged to local people, or was rescued from the Second Temple before its destruction or never existed to begin with.

Some of the Dead Sea Scrolls have been on interesting journeys. In 1948, a Syrian Orthodox archbishop known as Mar Samuel acquired four of the original seven scrolls from a Jerusalem shoemaker and part-time antiquity dealer, paying less than $100 for them. He then travelled to the United States and unsuccessfully offered them to a number of universities, including Yale. Finally, in 1954, he placed an advertisement in the business newspaper *The Wall Street Journal* – under the category 'Miscellaneous Items for Sale' – that read: 'Biblical Manuscripts dating back to at least 200 B.C. are for sale. This would be an ideal gift to an educational or religious institution by an individual or group.' Fortunately, Israeli archaeologist and statesman Yigael Yadin negotiated their purchase and brought the scrolls back to Jerusalem, where they remain to this day.

In 2017, researchers from the University of Haifa restored and deciphered one of the last untranslated scrolls. The university's Eshbal Ratson and Jonathan Ben-Dov spent one year reassembling the 60 fragments that make up the scroll. Deciphered from a band of coded text on parchment, the find provides insight into the community of people who wrote it and the 364-day calendar they would have used. The scroll names celebrations that indicate shifts in seasons and details two yearly religious events known from another Dead Sea Scroll. Only one more known scroll remains untranslated.

	c)	Trace the answer to the following questions. Choose NO MORE THAN THREE WORDS from the passage for each answer. i. Write the approximate timeline when the Roman army annihilated the scroll writers' community. ii. Mention the name of the book that is excluded from the dead sea scrolls. iii. Specify the category under which an advertisement was placed to sell the scrolls. iv. Name the position the person from Syria had when he bought the scrolls. v. Refer to the number of separate pieces the scrolls consist of.	[5x1=5]	
2.	a)	Write the correct form of the verbs in the bracket to complete the following conditional sentences as directed. i. If she ___ (know) about the traffic jam, she ___ (leave) earlier. (Use Third Conditional) ii. If the weather ___ (be) nice, we ___ (go) for a picnic. (Use First Conditional) iii. If you ___ (heat) metal, it ___ (expand) regardless of the type. (Use Zero Conditional) iv. If he ___ (not ignore) the warnings, he ___ (avoid) the accident last week. (Use Third Conditional) v. If I ___ (be) the president, I ___ (reduce) taxes. (Use Second Conditional) vi. If she ___ (prepare) the presentation tonight, she ___ (deliver) it confidently tomorrow. (Use First Conditional)	[6x0.5= 3]	CLO-1 *Level-C1*
	b)	Identify if the following sentences are Fragment, Run-on, Modifier, or Complete. If the sentence is complete, write "Complete Sentence"; if the sentence is Fragment, Run-on, or Modifier, then rewrite the sentence to fix the syntax error. i. The movie was interesting it had a great twist. ii. We decided to stay home because it was raining heavily. iii. Although he tried. iv. If you need help, don't hesitate to ask me. v. She likes tea he prefers coffee. vi. While walking to school. vii. I cooked dinner my brother set the table.	[7x1=7]	
3.	a)	One of your friends has asked you to be a partner in his new business. Compose a letter to your friend. In your letter: • Give your opinion of your friend's business idea • Tell him whether or not you have decided to accept the offer • Explain your reasons for this decision	[7]	CLO-4 *Level-C3*
	b)	Construct an essay on any one of the following topics: 1.The Causes of sleep deprivation among students and its effects on their academic life. 2. Everyone watches TV shows/series nowadays and it's hard to get up without finishing it. Some people think that binge watching TV-shows should be considered an addiction. Do you agree or disagree? Specify and Prove your stance.	[8]	

Daffodil International University
Faculty of Science & Information Technology
Department of Software Engineering
Final Examination, Summer-2025
Course Code: BNS 101, Course Title: Bangladesh Studies
Level: 1, Term: 1
Sections and Teachers:
A-D & I (DEH); E-H & N (DMR); J, K, L (MAA); M (SHS)

Exam Duration: 2 Hours **Marks: 40**

Answer ALL Questions

[The figures in the right margin indicate the full marks and corresponding course outcomes. All portions of each question must be answered sequentially.]

1	**Examine** how the Six-Point Movement transformed a regional autonomy demand into a full-fledged independence struggle.	8	CLO-3 L-4
2	**Interpret** the key points of ICT policy in Bangladesh. Can you find out the key challenges for gender equality in the ICT sector in Bangladesh?	8	CLO-4 L-5
3	**Predict** the future of Bangladesh economy in the light of the nature and recent trends of Bangladesh's economy.	8	CLO-3 L-4
4	**Apply** Lee's "Push-pull" model in the rural-urban migration in Bangladesh.	8	CLO-5 L-3
5	What types of challenges is Bangladesh facing in undertaking climate action as part of the Sustainable Development Goals (SDGs) under the UN? **Explain.**	8	CLO-4 L-5

Daffodil International University

Department of Software Engineering

Faculty of Science & Information Technology

Final Examination; Summer 2025

Course Code: SE 113; Course Title: Introduction to Software Engineering

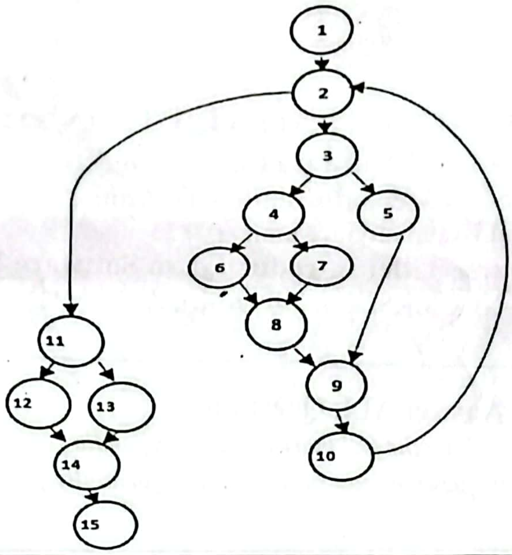
Sections & Teachers: 45 (A-N) & MKS, RM, MIF, KMH, ZNM, MMN, FMD, MTM

Time: 2:00 Hrs. **Marks: 40**

Answer ALL Questions

[The figures in the right margin indicate the full marks and corresponding course outcomes. All portions of each question must be answered sequentially.]

1.	a)	Interpret these problems and classify them under the appropriate software risk categories 1. Developers are confused due to conflicting client requirements. 2. Budget was exceeded after purchasing tools that did not integrate properly. 3. Project timeline was missed due to lengthy training of newly hired staff. 4. Final product is buggy and lacks documentation for future updates. 5. Frequent requirement changes forced architectural redesign. 6. Testing was skipped to meet delivery deadlines. 7. Project delivery was delayed due to poor estimation of team capabilities. 8. Project suffered due to sudden resignation of key developers. 9. Poorly defined requirements led to major rework in development. 10. Developers misunderstood user needs due to lack of formal documents.	**Marks - 5**	**CLO-1 Level-2**
	b)	**Discuss** the advantages of applying Gray Box Testing that the tester may gain. Additionally, **Distinguish** the key differences between Black Box Testing and White Box Testing in terms of their respective approaches within the software development process.	**Marks – 2+3**	
2.		Daffodil International University has an online student portal where students can log in to view their class routines, submit assignments, check exam results, and communicate with course teachers. Teachers can upload course materials, give assignments, and provide feedback. Admin can manage user accounts and system access.		
	a)	**Derive** a Use Case Diagram identifying major actors and actions of the system.	**Marks –5**	
	b)	**Produce** a description of the use case "Submit Assignment" with appropriate course of action.	**Marks - 4**	
	c)	**Figure** the Cyclomatic complexity of the following flow chart. Verify your answer using Graph matrix theory.	**Marks - 6**	**CLO-2 Level-3**



3. a) **Visualize** the network diagram from the above table. **Observe** all the possible path and their duration to compute the critical path of the project. **Marks - 6**

Activity	Immediate Predecessor	Estimated Duration
A	None	3
B	A	4
C	A	6
D	B	5
E	C	9
F	D,E	7
G	C	5
I	G	6
H	F,I	5
J	H	2

CLO-3 Level-2

b) **Elaborate** the contribution of each phases of the Project Management Process toward the successful completion of the software product. **Marks - 3**

c) Consider a software project for an e-commerce system. The Lines of Code (LOC) of major software functions identified are: **Marks - 6**

For Product listings management (LOC):	For Order processing (LOC):	For Payment handling (LOC):	For Delivery tracking (LOC):
S_{opt}: 3200 S_m : 4700 S_{pess}:7500	S_{opt}: 3600 S_m : 5200 S_{pess} 8200	S_{opt}: 2400 S_m : 3600 S_{pess}: 5700	S_{opt}: 2100 S_m .: 2900 S_{pess}:4500

In this software project, the productivity rate is estimated at 4200 lines of code (LOC) per person-month, and the development cost is $6500 per person-month. **Compute** the following:

i. Number of person-months required for development.
ii. Total cost of software development.
iii. Cost per LOC.

Daffodil International University

Department of Software Engineering

Faculty of Science & Information Technology

Final Examination, Summer 2025

Course Code: SE 111; Course Title: Computer Fundamentals

Sections & Teachers: STA(A,B), SI(C,D, E), MH(F, G), MIF(H), KFH (I), SHN (J), AAS (K), SAR (L), MS (M), DDK (N)

Time: 2:00 Hrs **Marks: 40**

Answer <u>ALL</u> Questions

[The figures in the right margin indicate the full marks and corresponding course outcomes. All portions of each question must be answered sequentially.]

1. ***a)*** Fazla Rabby Raihan is a sports coordinator at Software Engineering Department, DIU. He is calculating scores for an inter-batch football tournament. Each team earns **3 points** for a win, **1 point** for a draw, and **0 points** for a loss. He totals the points for each team and applies the ranking rules as given below: **Marks-15** **CLO-3** ***C-3***

- If points ≥ 15: Award "Champion" trophy.
- If 10 ≤ points < 15: Award "Runner-up" trophy.
- If points < 10: Give participation certificates only.

So, **construct** an algorithm to help Fazla Rabby Raihan to print points after calculating with award and **draw** the flowchart from your algorithm.

b) **Detect** errors in the following C code, **update** it after correcting the errors, and then write down the corrected version. Also, **show** the expected output when the **input is: 20 19 21** **Marks-5**

```
#include <studio,h>
int main {
   int num1 num2 num3
   printf(Enter three numbers: )
   scanf("%d %d %d", &num1 &num2 &num3)
   if (num1 <= num2 && num1 <= num3)
   {
     printf("Smallest number is: %d\n", num1)
   }
   else if (num2 <= num1 & num2 <= num3)
   {
     printf("Smallest number is: %d\n", num2)
   }
   else
     printf("Smallest number is: %d\n", num3)
   return 0
   }
```

	c)	Ashik Ahmed Izaz and Sefatullah set up their gaming consoles so that messages travel in a loop among all their friends. When Ashik sends a game invite, it passes through each friend one by one until it reaches Sefatullah, who then replies back the same way. If one friend's console goes offline, the message can't complete the loop, and the game invite is lost until the connection is fixed. Due to this problem they decided to connect every gaming console directly with all other consoles. This way, Ashik can send a game invite straight to Sefatullah without relying on anyone else. **Determine** the network topology and the characteristics they used before and after connecting every gaming console directly with all other consoles with proper diagram. ring, mesh	**Marks-5**	
	d)	At the DIU Library, students often borrow books through the online system. When a student requests a book, the system updates both the borrowing record and the book's availability at the same time, ensuring the process is either fully completed or not done at all. If a failure occurs midway, all changes are automatically reversed, keeping the data accurate. Even during peak times, when many students request the same textbook, each transaction is handled independently, preventing conflicts. Once a borrowing is confirmed, the record is stored permanently, unaffected by power outages or crashes, ensuring the library's database remains consistent and reliable. **Determine** the database transaction properties shown in the scenario and **demonstrate** their role in handling the process of DIU library.	**Marks-5**	
2.	*a)*	Abdul Hai Zebon was logged into his online banking account when he received an email claiming he had won a prize. Without realizing it, he clicked a link that automatically submitted a request to transfer money from his account to an unknown recipient. The request appeared legitimate because he was already authenticated on the bank's website. This unwanted action occurred without his consent. **Detect** what type of vulnerability mentioned in the above scenario and **explain** what should be followed by Mr. Zebon to prevent any type of vulnerabilities. CSRF	**Marks-5**	**CLO-4** *C-4*
	b)	At a university portal, Soikot creates his password to access his account. Instead of storing the actual passwords, the system converts the password into a fixed-length code that cannot be reversed. Even if someone accesses the database, they cannot discover the original password. Also, when soikot submit important assignment files, the system generates a unique code for each file. Later, when the files are accessed or downloaded, the system checks the code to ensure the files have not been altered, protecting both security and data integrity. **Identify** the specific technique mentioned in the scenario. **Explain** the importance of this technique in real life. Hashing	**Marks-5**	